UBUNTU_0 by Blága Máté

This is a common guide for ubuntu configurations

Network Settings

Via Netplan

The Config file is in `/etc/netplan/01-netcfg.yaml`

Set Static IP

```
network:
  version: 2
  ethernets:
    <INTERFACE NAME>:
      dhcp4: false
      addresses:
        - <IP Address>/<SM>
        - <IPv6 Address>/<prefix>
      routes:
        - to: default
          via: <Default Gatway>
        - to: default
          via: <IPv6 Address>
```

Set DHCP

```
network:
  version: 2
  ethernets:
    <INTERFACE NAME>:
      dhcp4: true
      dhcp6: true
```

Validate Netplan Configuration

```
 sudo netplan generate
```

Apply Netplan config

```
sudo netplan apply
```

User Management

Add User with home directory

```
sudo useradd -m <USERNAME>
```

Add Group

```
sudo groupadd <GROUPNAME>
```

Add User to Group

```
sudo usermod -aG <GROUPNAME> <USERNAME>
```

Remove User from Group

```
sudo gpasswd -d <USERNAME> <GROUPNAME>
```

Delete User

```
sudo deluser <USERNAME>
```

Delete Group

```
sudo groupdel <GROUPNAME>
```

Change User Password

```
sudo passwd <USERNAME>
```

Add User to Sudo Group

```
sudo usermod -aG sudo <USERNAME>
sudo visudo # Add the following line at the end of the file
<USERNAME> ALL=(ALL) NOPASSWD:ALL
```

File Permissions

Permission Representation

Num	Per	Bin
7	rwx	111
6	rw-	110
5	r-x	101
4	r–	100
3	-wx	011
2	-w-	010
1	–x	001
0	—	000

Change File Owner

```
sudo chown <USERNAME>:<GROUPNAME> <FILE>
```

Change File Permissions

```
sudo chmod <PERMISSIONS> <FILE/DIRECTORY>
```

Change File Permissions Recursively

```
sudo chmod -R <PERMISSIONS> <DIRECTORY>
```

ACL Permissions

0. Step is to install ACL package

```
sudo apt install acl
```

Set ACL permissions for a user/group

```
sudo setfacl -m <u/g>:<USERNAME>:<PERMISSIONS> <FILE/DIRECTORY>
```

View ACL permissions for a file/directory

```
getfacl <FILE/DIRECTORY>
```

Remove ACL permissions for a user/group

```
sudo setfacl -x <u/g>:<USERNAME> <FILE/DIRECTORY>
```

Set quota

1. Install quota package

```
sudo apt install quota
```

2. Edit `/etc/fstab` and add usrquota and grpquota options to the desired partition

```
/dev/sda1 / ext4 defaults,usrquota,grpquota 0 1
```

3. Remount the partition

```
systemctl daemon-reload
sudo mount -o remount /
```

4. Create quota files

```
sudo quotacheck -ugm /
```

5. Enable quota

```
sudo quotaon -v /
```

6. Set quota for a user

```
sudo setquota -u <USERNAME> <SOFT LIMIT> <HARD LIMIT> 0 0 /
```

7. View quota for a user

```
sudo quota -u <USERNAME>
```

8. Remove quota for a user

```
sudo setquota -u <USERNAME> 0 0 0 0 /
```

TAR Command

Create a tar archive

```
tar -cvf <ARCHIVE_NAME>.tar <FILE/DIRECTORY>
```

Extract a tar archive

```
tar -xvf <ARCHIVE_NAME>.tar -C <DESTINATION_DIRECTORY>
```

Compression methods

Option	Compression Method	File Extension
-z	gzip	.tar.gz
-j	bzip2	.tar.bz2
-J	xz	.tar.xz
–zstd	zstd	.tar.zst

DHCP Server Configuration

1. Install DHCP server package

```
sudo apt install isc-dhcp-server
```

2. Edit DHCP server configuration file

```
sudo nano /etc/dhcp/dhcpd.conf
```

3. Add the following lines to the configuration file

```
subnet <SUBNET> netmask <NETMASK> {
  range <START_IP> <END_IP>;
  option routers <DEFAULT_GATEWAY>;
  option domain-name-servers <DNS_SERVER>;
}
```

4. Restart DHCP server

```
sudo systemctl restart isc-dhcp-server
```

DNS Server Configuration

1. Install DNS server package

```
sudo apt install dnsmasq
```

2. Edit DNS server configuration file

```
sudo nano /etc/dnsmasq.conf
```

3. Change the following lines in the configuration file

```
listen-address=<IP_ADDRESS>
```

4. Stop systemd-resolved service and change the dns server

```
sudo systemctl stop systemd-resolved
sudo systemctl disable systemd-resolved
echo "nameserver <IP_ADDRESS>" | sudo tee /etc/resolv.conf
```

5. Restart DNS server

```
sudo systemctl restart dnsmasq
```

6. Add custom DNS records

```
sudo nano /etc/hosts
```

Add the following lines to the hosts file

```
<IP_ADDRESS> <HOSTNAME>
```

Enable ipv4 forwarding

1. Edit sysctl configuration file

```
sudo nano /etc/sysctl.conf
```

2. Add the following line to the configuration file

```
net.ipv4.ip_forward=1
```

3. Apply the changes

```
sudo sysctl -p
```

Apache Web Server Configuration

1. Install Apache web server package

```
sudo apt install apache2
```

2. Create a new virtual host configuration file

```
sudo nano /etc/apache2/sites-available/<DOMAIN_NAME>.conf
```

3. Add the following lines to the configuration file

```
<VirtualHost *:80>
    ServerName <DOMAIN_NAME>
    DocumentRoot /var/www/<DOMAIN_NAME>
    <Directory /var/www/<DOMAIN_NAME>>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

4. Enable the new virtual host

```
sudo a2ensite <DOMAIN_NAME>.conf
```

5. Disable the default virtual host

```
sudo a2dissite 000-default.conf
```

6. Restart Apache web server

```
sudo systemctl restart apache2
```

PHP Configuration

1. Install PHP package

```
sudo apt install php libapache2-mod-php
```

2. Create a new PHP file in the document root

```
sudo nano /var/www/<DOMAIN_NAME>/index.php
```

3. Add the following lines to the PHP file

```
<?php
phpinfo();
?>
```

4. Restart Apache web server

```
sudo systemctl restart apache2
```

MySQL Database Server Configuration

1. Install MySQL database server package

```
sudo apt install mysql-server
```

2. Secure MySQL installation

```
sudo mysql_secure_installation
```

3. Log in to MySQL shell

```
sudo mysql -u root -p
```

4. Create a new database

```
CREATE DATABASE <DATABASE_NAME>;
```

5. Create a new user and grant privileges

```
CREATE USER '<USERNAME>'@'localhost' IDENTIFIED BY '<PASSWORD>';
GRANT ALL PRIVILEGES ON <DATABASE_NAME>.* TO '<USERNAME>'@'localhost';
FLUSH PRIVILEGES;
```

6. Exit MySQL shell

```
EXIT;
```

7. Restart MySQL database server

```
sudo systemctl restart mysql
```

SSH Server Configuration

1. Install SSH server package

```
sudo apt install openssh-server
```

2. Edit SSH server configuration file

```
sudo nano /etc/ssh/sshd_config
```

3. Change the following lines in the configuration file

```
Port <PORT_NUMBER>
PermitRootLogin no
PasswordAuthentication no
```

4. Restart SSH server

```
sudo systemctl restart ssh
```

Fail2Ban Configuration

1. Install Fail2Ban package

```
sudo apt install fail2ban
```

2. Create a new jail configuration file

```
sudo nano /etc/fail2ban/jail.local
```

3. Add the following lines to the configuration file

```
[sshd]
enabled = true
port = <PORT_NUMBER>
filter = sshd
logpath = /var/log/auth.log
maxretry = 5
```

4. Restart Fail2Ban service

```
sudo systemctl restart fail2ban
```

Useful Ports

Service	Port Number	Protocol
SSH	22	TCP
HTTP	80	TCP
HTTPS	443	TCP
HTTP Proxy	8080	TCP
HTTPS Proxy	8443	TCP
MySQL	3306	TCP
PostgreSQL	5432	TCP
DNS	53	TCP/UDP
DHCP	67	UDP
NTP	123	UDP
SNMP	161	UDP
SNMP Trap	162	UDP
Samba	445	TCP
FTP	21	TCP
FTP (Data)	20	TCP
SFTP/SSH	22	TCP
TFTP	69	UDP
SMTP	25	TCP
SMTP (SSL)	465	TCP
POP3	110	TCP
POP3 (SSL)	995	TCP
IMAP	143	TCP
IMAP (SSL)	993	TCP

IPTables Configuration

Install IPTables package

```
sudo apt install iptables
```

Check current IPTables rules

```
sudo iptables -L -v
```

Allow incoming port

```
sudo iptables -A INPUT -p <tcp/udp> --dport <PORT_NUMBER> -j ACCEPT
```

Drop all other incoming connections

```
sudo iptables -A INPUT -j DROP
```

Save IPTables rules

```
sudo iptables-save | sudo tee /etc/iptables/rules.v4
```

Load IPTables rules on boot

```
sudo nano /etc/rc.local
```

Add the following lines to the rc.local file

```
#!/bin/bash
iptables-restore < /etc/iptables/rules.v4
exit 0
```

Dont forget to make rc.local file executable

```
sudo chmod +x /etc/rc.local
```

UFW Configuration

Install UFW package

```
sudo apt install ufw
```

Allow a specific port

```
sudo ufw allow <PORT_NUMBER>/<tcp/udp>
```

Allow a specific service

```
sudo ufw allow <SERVICE_NAME>
```

Enable UFW

```
sudo ufw enable
```

Check UFW status

```
sudo ufw status verbose
```

Restart UFW

```
sudo ufw reload
```

SAMBA Server Configuration

1. Install SAMBA package

```
sudo apt install samba
```

2. Edit SAMBA configuration file

```
sudo nano /etc/samba/smb.conf
```

3. Add the following lines to the configuration file

```
[SHARE_NAME]
   path = /path/to/share
   available = yes
   valid users = <USERNAME>
```

```
read only = no
browsable = yes
public = yes
writable = yes
```

4. Create a new SAMBA user

```
sudo smbpasswd -a <USERNAME>
```

5. Restart SAMBA service

```
sudo systemctl restart smbd
```

Sambe Client Configuration

Install SAMBA client package

```
sudo apt install samba-client
```

Mount a SAMBA share

```
sudo mount -t cifs //<SERVER_IP>/<SHARE_NAME> /mnt/samba -o
username=<USERNAME>,password=<PASSWORD>
```

CRON Job Configuration

Install CRON package

```
sudo apt install cron
```

Edit CRON jobs

```
sudo crontab -e
```

Add a new CRON job

```
* * * * * /path/to/script.sh
```

Restart CRON service

```
sudo systemctl restart cron
```

Cron syntax

Field	Allowed Values	Special Characters
Minute	0-59	* / , -
Hour	0-23	* / , -
Day of Month	1-31	* / , -
Month	1-12 or Jan-Dec	* / , -
Day of Week	0-7 or Sun-Sat (0 and 7 are Sunday)	* / , -

Example CRON job to run a script every 5 minutes

```
*/5 * * * * /path/to/script.sh
```

Rsyslog Configuration

Install Rsyslog package

```
sudo apt install rsyslog
```

Check Rsyslog

```
sudo cat /var/log/syslog
```

Edit Rsyslog configuration file

```
echo "*.* /var/log/all.log" | sudo tee -a /etc/rsyslog.conf
```

Restart Rsyslog service

```
sudo systemctl restart rsyslog
```

Fdisk Command - Disk Partitioning

List all disks and partitions

```
sudo fdisk -l
```

Create a new partition

```
sudo fdisk /dev/sdX
```

Follow the interactive prompts to create a new partition. After creating the partition, you need to format it and mount it to use it.

Format the new partition for ext4 filesystem

```
sudo mkfs.ext4 -L <Lable Name> /dev/sdX1
```

Create a mount point and mount the new partition

```
sudo mkdir /mnt/new_partition
sudo mount /dev/sdX1 /mnt/new_partition
```

Add the new partition to /etc/fstab for automatic mounting on boot by Label

```
echo "LABEL=<Lable Name> /mnt/new_partition ext4 defaults 0 2" | sudo tee -a
/etc/fstab
```

Remount by Fstab settings

```
systemctl daemon-reload
sudo mount -a
```

Delete a partition

```
sudo fdisk /dev/sdX
```

Follow the interactive prompts to delete the desired partition. Be careful when deleting partitions, as it will result in data loss.

Resize a partition

```
sudo fdisk /dev/sdX
```

Follow the interactive prompts to delete the existing partition and create a new partition with the desired size. After resizing the partition, you need to resize the filesystem to use the new size.

Rsnapshot Configuration - Backup Tool

TODO

SCRIPTING

Set Variables

```
VARIABLE_NAME="Value"
```

Use Variables

```
echo "The value of the variable is: $VARIABLE_NAME"
```

Unset Variables

```
unset VARIABLE_NAME
```

Read User Input

```
read -p "Enter your name: " NAME
echo "Hello, $NAME!"
```

IF statement

```
if [ <CONDITION> ]; then
    # Commands to execute if condition is true
elif [ <ANOTHER_CONDITION> ]; then
    # Commands to execute if another condition is true
else
    # Commands to execute if all conditions are false
fi
```

FOREACH loop

```
for <VARIABLE> in <LIST>; do
    # Commands to execute for each item in the list
done
```

FOR loop

```
for (( i=0; i < 10; i+=1 )); do
    # Commands to execute for each iteration
done
```

WHILE loop

```
while [ <CONDITION> ]; do
    # Commands to execute while condition is true
done
```